

Zomato Restaurant Data Analysis

Ratings, Cuisines, Location & Pricing Insights

Dataset: zomato.csv | 51,717 clean records analyzed

Key Findings

- Of 56,252 raw rows, 51,717 were usable after removing 4,535 rows corrupted by broken CSV quoting (8.1% of the file).
- Modern Indian is the top-rated major cuisine (avg 4.31/5, 145 listings), while Hyderabadi rates lowest among common cuisines (avg 3.52/5).
- North Indian, Chinese and South Indian are the three most widely offered cuisines by listing volume.
- Whitefield has the most restaurants (631), making it the single biggest hotspot in the dataset.
- Cost for two correlates positively with rating ($r = 0.39$) — pricier restaurants trend modestly higher-rated, but it's a weak relationship, not a strong driver.
- Votes correlate more strongly with rating ($r = 0.43$) than cost does — popularity/engagement tracks rating more than price does.
- Restaurants offering online ordering and table booking both show visibly higher median ratings than those that don't, in the boxplots above.
- A Random Forest model predicting rating from votes, cost, location, cuisine, rest type and listing type achieves $R^2 = 0.68$ (vs. 0.33 for plain Linear Regression). 'votes' alone accounts for 62% of feature importance — engagement/popularity predicts rating far more than cuisine, price, or location do.

Predictive Model: What Drives Rating?

index	Model	R2	MAE	RMSE
0	Linear Regression	0.33	0.263	0.335
1	Random Forest	0.685	0.16	0.23

Top 10 Feature Importances (Random Forest):

votes	0.618
cost_for_two	0.069
cuisine_Chinese	0.037
cuisine_North Indian	0.023
book_table	0.022
cuisine_Desserts	0.020
cuisine_Fast Food	0.011
rest_type_grp_Quick Bites	0.010
online_order	0.010
rest_type_grp_Casual Dining	0.009

Recommendations for Alfido Tech-style Platform

1. Localized 'Hotspot' Partnership Packages

Concentrate merchant-acquisition and sponsored-placement efforts on the top restaurant-dense neighborhoods (e.g. Whitefield, BTM, HSR, Marathahalli, Electronic City). These areas have the deepest restaurant supply and the most price/rating variance — the highest-value real estate for featured listings and neighborhood-level ad packages.

2. Cuisine-Based Content & Curated Collections

Build recurring editorial content ('Best Modern Indian Under ₹500', 'Top-Rated Malaysian Spots') around the highest-rated, highest-demand cuisines identified in the cuisine analysis. This drives organic search traffic and gives restaurants a reason to keep their listings current.

3. Reward Online-Ordering & Table-Booking Adoption

Since restaurants offering online ordering and table booking show higher median ratings, incentivize non-adopting restaurants (discounted onboarding fees, visibility boosts) to enable these features — it's a lever that correlates with better customer experience, not just convenience.

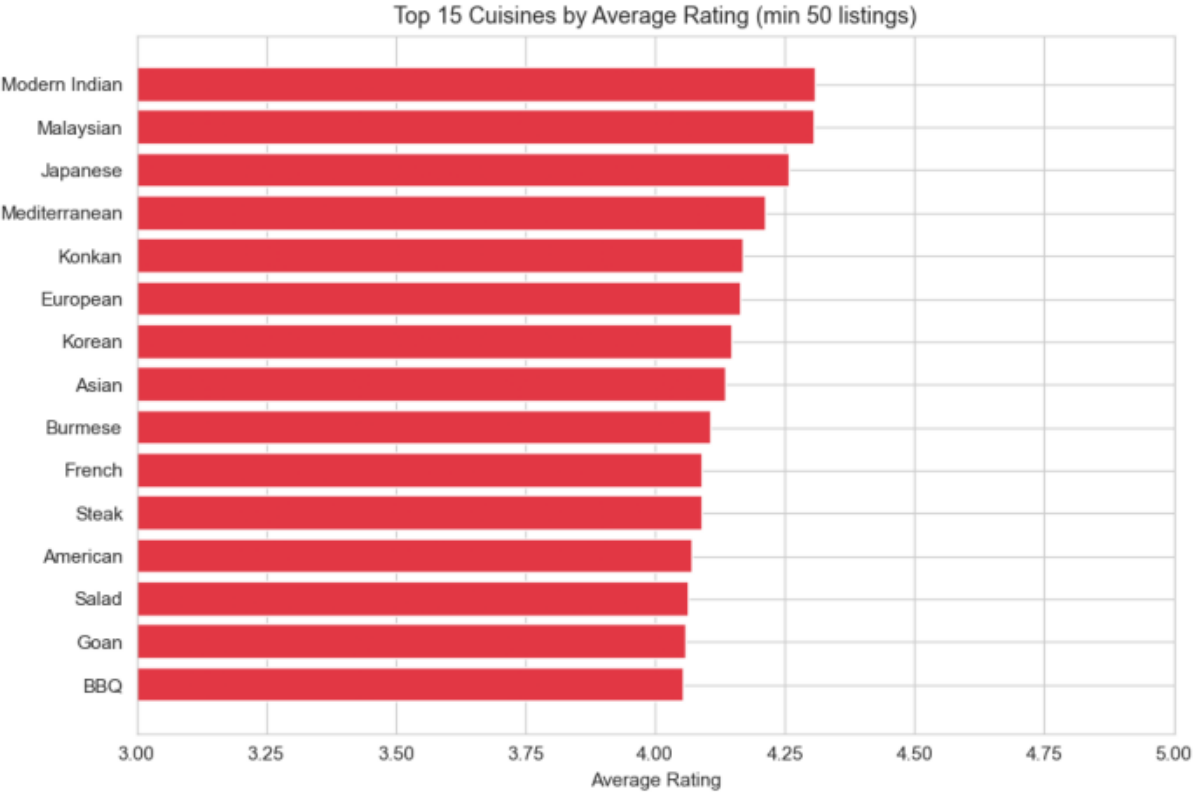
4. Price-Tier Discovery Filters, Not Just Price Sorting

Since cost only weakly predicts rating, help users find 'hidden gem' budget restaurants with high ratings rather than assuming higher price = better quality. A 'Best Value' filter (high rating, low cost bucket) would differentiate the platform from pure price-sort competitors.

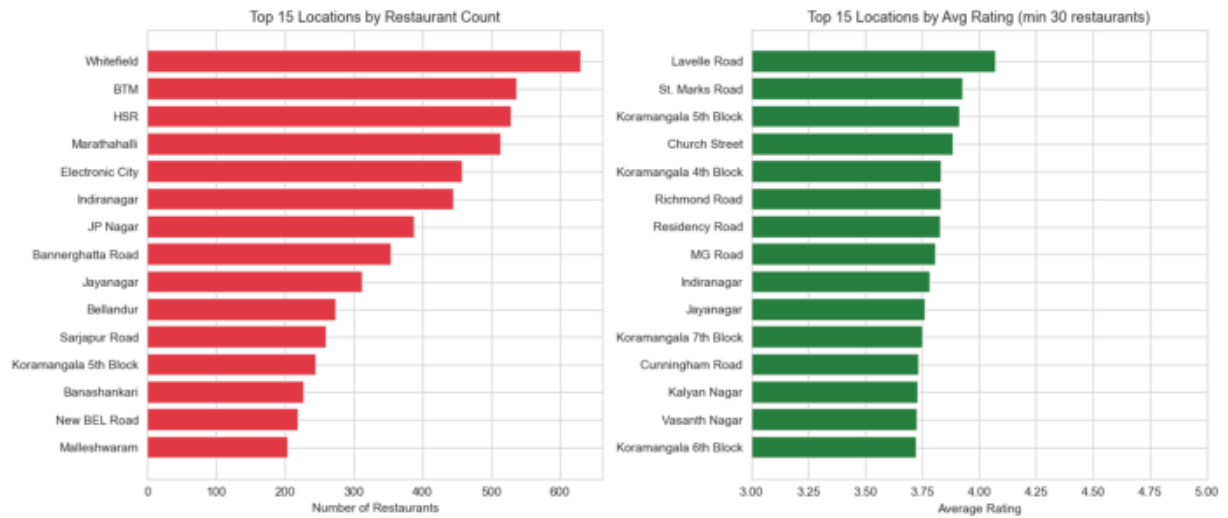
5. Fix Structured Review Capture at the Data Layer

This raw dataset shows what happens when free-text reviews aren't captured in clean, structured fields — 8% of records were unusable due to quoting/escaping corruption. A platform should store review text, sub-ratings (taste/service/ambience), and dish mentions as separate structured fields from day one — enabling exactly the sentiment analysis and dish-level wordclouds this dataset couldn't support.

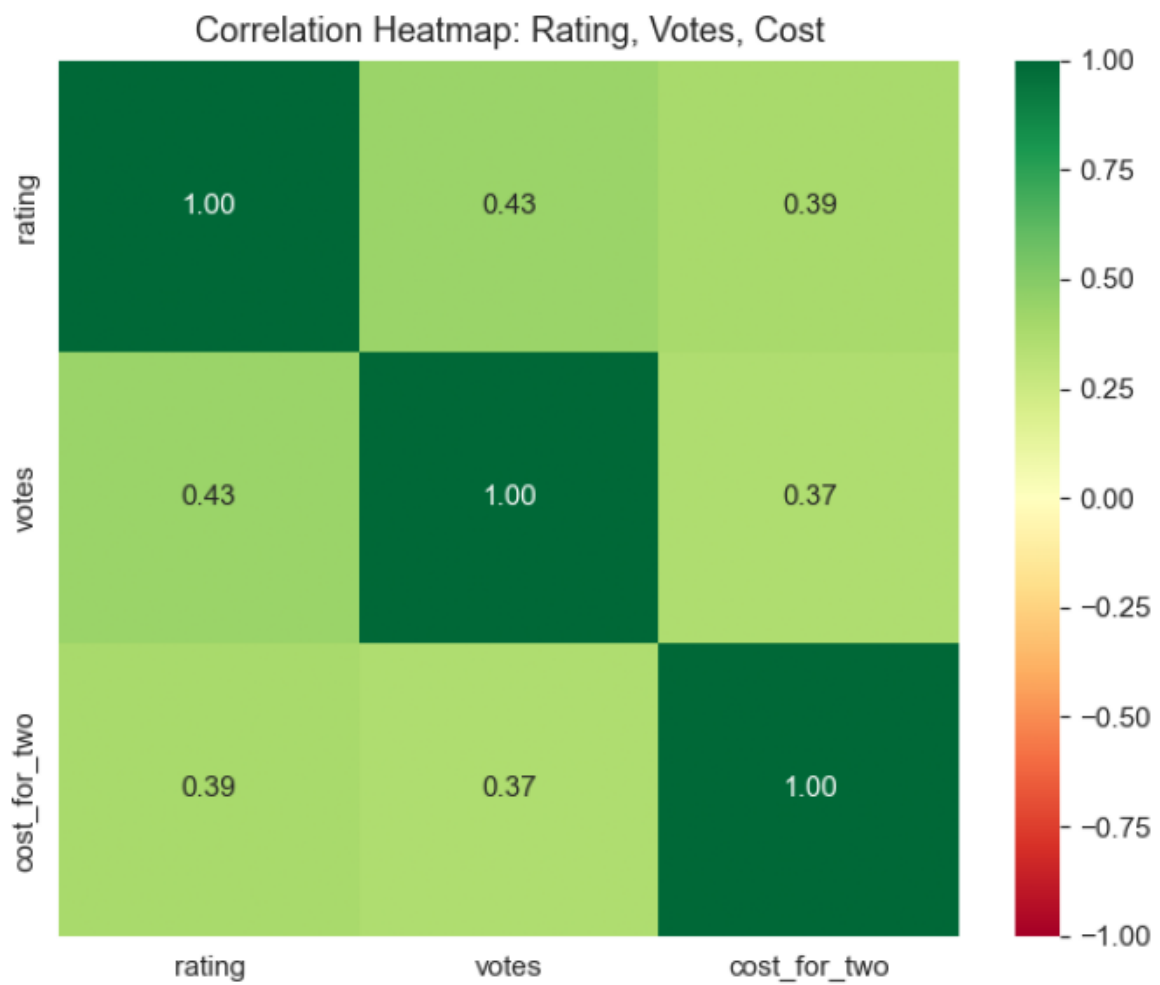
Cuisine vs Rating



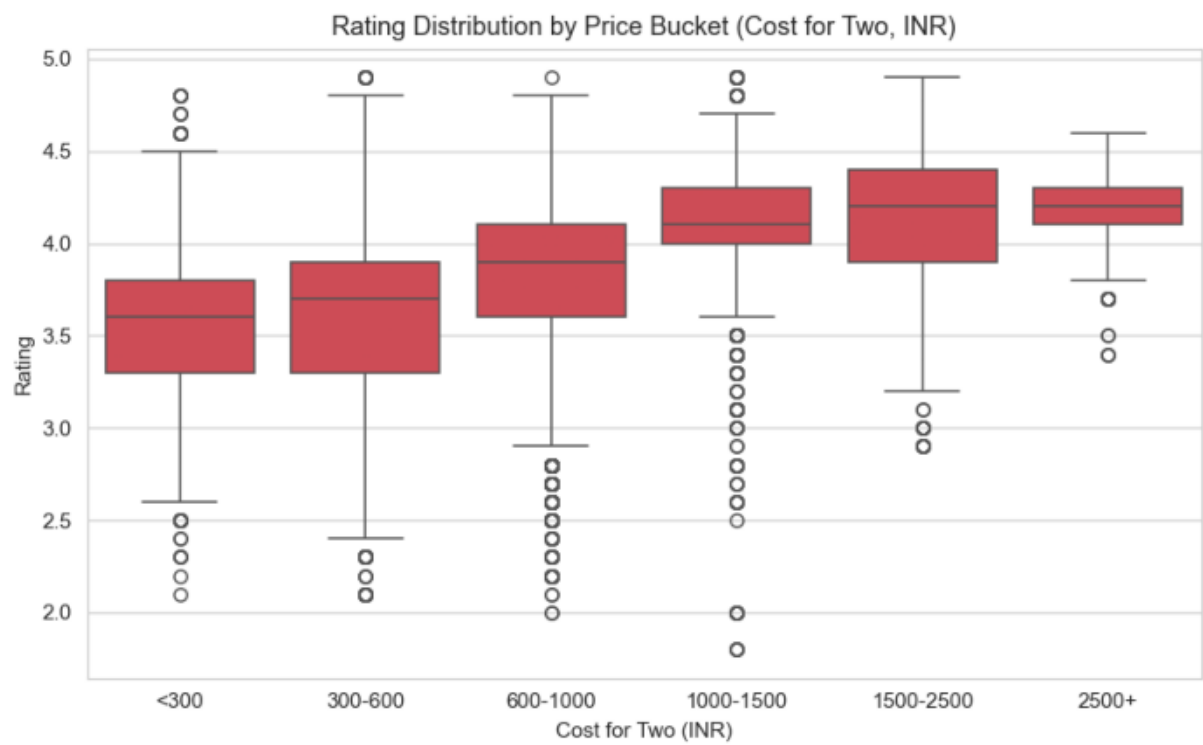
Location Hotspots



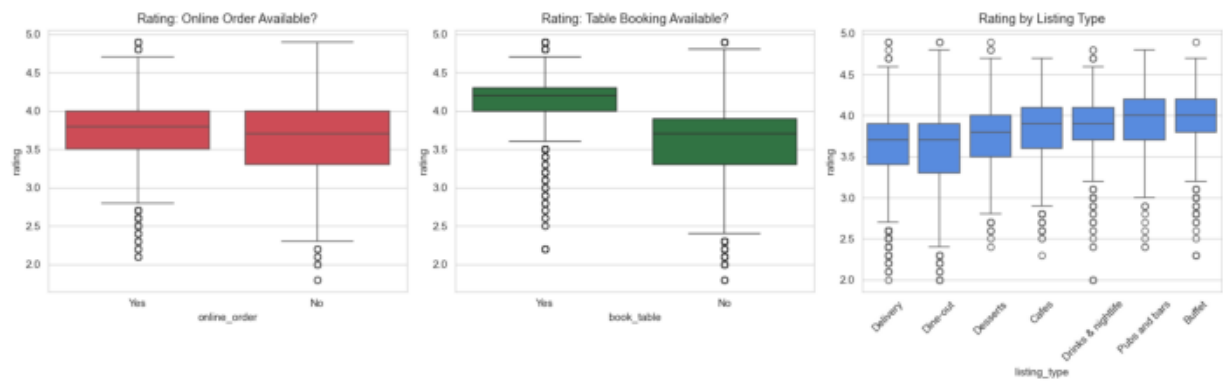
Correlation Heatmap



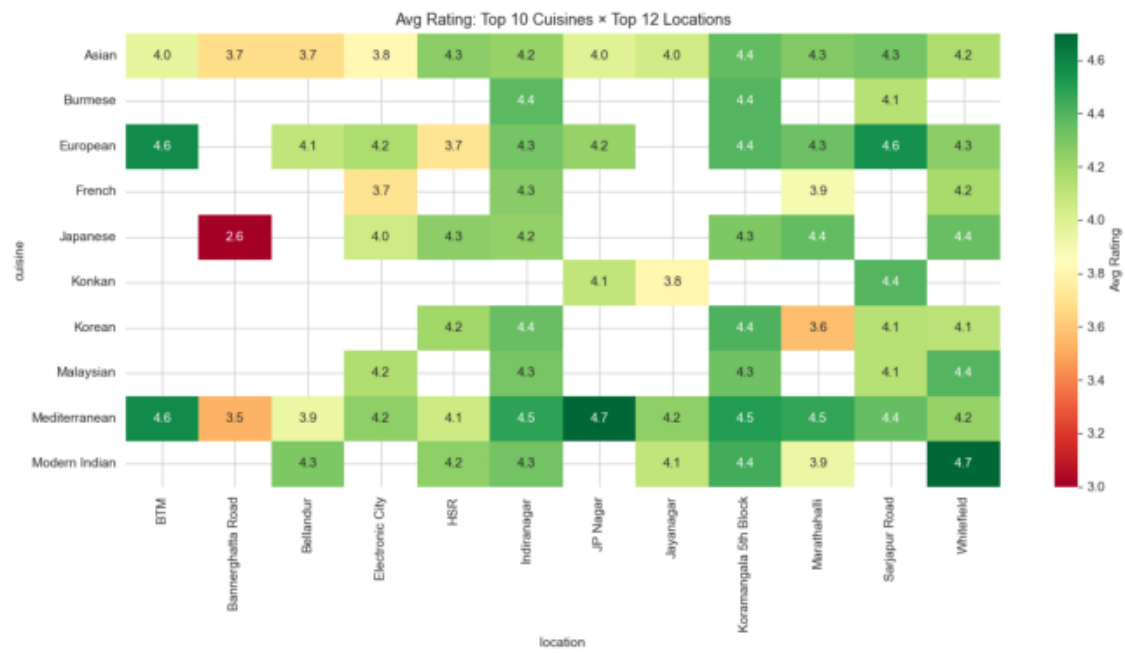
Price vs Rating



Other Rating Factors



Cuisine x Location Heatmap



Cuisine & Dish Wordclouds

